

A disproof of Conjecture 00000000981

No compact operator has point spectrum dense in the unit disk

1 The conjecture

Conjecture 00000000981 reads:

Conjecture: There exists a compact operator T with $\sigma_p(T)$ dense in the unit disk, yet every invariant subspace of T contains an eigenvector (density-eigenvector tension).

This is an existence claim with two conditions on T . We show that the first condition alone is impossible: *no compact operator on any complex normed space (in particular any Banach or Hilbert space) has point spectrum dense in the open unit disk.* Hence no T satisfies both conditions, however the second condition is read.

2 Definitions and conventions

- E is a complex normed space (no completeness is needed). A linear map $T : E \rightarrow E$ is *compact* if it maps some neighbourhood of 0 into a compact set (Mathlib `IsCompactOperator`); such a map sends bounded sets into compact sets. The main statement is given both for linear maps and for bounded operators `E → L[C] E`.
- The point spectrum is $\sigma_p(T) = \{\mu \in \mathbb{C} : \exists x \neq 0, Tx = \mu x\}$ (Lean: `pointSpectrum`).
- “ $\sigma_p(T)$ dense in the unit disk” is read as $D \subseteq \overline{\sigma_p(T)}$ with $D = \{|\mu| < 1\}$. This is the weakest reading: density of $\sigma_p(T) \cap D$ in D , or density in the closed disk, both imply it (the closed-disk version is `not_closedBall_subset_closure_pointSpectrum`).
- “Every invariant subspace contains an eigenvector” is formalized as: every nonzero closed T -invariant subspace contains some $x \neq 0$ with $Tx = \mu x$ (Lean: `EveryInvariantSubspaceHasEigenvector`). The proof never uses this condition.

3 The proof

Lemma 1 (finitely many large eigenvalues). *If T is compact and $\delta > 0$, then $\{\mu \in \sigma_p(T) : |\mu| \geq \delta\}$ is finite.*

Proof. Suppose not. Choose distinct $\lambda_0, \lambda_1, \dots$ in this set and $v_k \neq 0$ with $Tv_k = \lambda_k v_k$. Eigenvectors for distinct eigenvalues are linearly independent (Mathlib `Module.End.eigenvectors_linearIndependent`). Put $M_n = \text{span}(v_0, \dots, v_{n-1})$: finite-dimensional, hence closed; $M_n \subseteq M_{n+1}$; $v_n \in M_{n+1} \setminus M_n$; $T(M_n) \subseteq M_n$; and $(T - \lambda_n)(M_{n+1}) \subseteq M_n$, since $(T - \lambda_n)v_k = (\lambda_k - \lambda_n)v_k$.

By the Riesz lemma (Mathlib `riesz_lemma_of_lt_one`) applied in M_{n+1} to its proper closed subspace M_n , there is $y_n \in M_{n+1}$ with $\|y_n\| = 1$ and $\|y_n - z\| \geq 1/2$ for all $z \in M_n$. For $m < n$ let $z = \lambda_n^{-1}(\lambda_n y_n - T y_n + T y_m) \in M_n$; then $T y_n - T y_m = \lambda_n(y_n - z)$, so $\|T y_n - T y_m\| \geq |\lambda_n|/2 \geq \delta/2$. But the y_n lie in the closed unit ball, so the $T y_n$ lie in a compact set and have a convergent, hence Cauchy, subsequence, contradicting $\|T y_{\varphi(N+1)} - T y_{\varphi(N)}\| \geq \delta/2$. (Lean: `finite_eigenvalues_ge`.) $\square$

Theorem 2. *If T is compact, then $D \not\subseteq \overline{\sigma_p(T)}$.*

Proof. Let $F = \{\mu \in \sigma_p(T) : |\mu| \geq 1/2\}$, finite by Lemma 1. The real segment $(1/2, 1)$ is infinite, so it contains some $t \notin F$. The set $U = \{|\mu| > 1/2\} \setminus F$ is open and contains t . If $t \in D \subseteq \overline{\sigma_p(T)}$, then U meets $\sigma_p(T)$ at some μ ; but then $\mu \in F$, a contradiction. (Lean: `not_ball_subset_closure_pointSpectrum`.) $\square$

Theorem 3. *For every complex normed space E there is no compact T (linear map, or bounded operator) on E with $D \subseteq \sigma_p(T)$ and every nonzero closed invariant subspace containing an eigenvector. The conjecture is false.*

4 Lean formalization

File `lean/Conjecture981/Basic.lean` (namespace `C981`, Lean 4 + Mathlib, only `import Mathlib`). Main theorems:

- (ASCII rendering) `conjecture_false (E) : not (exists T : E ->[C] E, IsCompactOperator T and ball 0 1 subset closure (pointSpectrum T) and EveryInvariantSubspaceHasEigenvector T);`
- `conjecture_false_clm`: the same for bounded operators $T : E \rightarrow_{\mathbb{C}} E$;
- `finite_eigenvalues_ge`, `not_ball_subset_closure_pointSpectrum`, `not_closedBall_subset_closure_pointSpectrum`.

Here E is any type with `NormedAddCommGroup E` and `NormedSpace C E`; Banach and Hilbert spaces are special cases. Axiom audit: `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`.

Remark 4. Not covered: real scalars. Over $\mathbb{R}$ eigenvalues are real numbers and cannot be dense in a disk of $\mathbb{C}$. We read “unit disk” as the disk, not the unit circle, as both language versions say.